

**Institute of Mathematical Sciences
University of Malaya**

**SIM1001 Basic Mathematics
SIX1013 Fundamentals of Advanced Mathematics**

TUTORIAL 2 (Part II)

1. Find the inverse of each of the following matrices:

(a) $A = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}, \quad c \in \mathbb{R}$ (b) $B = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad \theta \in \mathbb{R}.$

2. Let λ be a non-zero scalar, show that $\begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{\lambda} & -\left(\frac{1}{\lambda}\right)^2 & \left(\frac{1}{\lambda}\right)^3 \\ 0 & \frac{1}{\lambda} & -\left(\frac{1}{\lambda}\right)^2 \\ 0 & 0 & \frac{1}{\lambda} \end{bmatrix}.$

3. Let $A, B \in M_n(\mathbb{R})$ be invertible matrices. Show that

(a) λA is invertible and $(\lambda A)^{-1} = \frac{1}{\lambda} A^{-1}$ for a nonzero scalar λ .

(b) A^{-1} is invertible, and $(A^{-1})^{-1} = A$.

(c) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

4. Let $A, B \in M_n(\mathbb{R})$.

(a) If A and B are nonsingular, is $A + B$ also nonsingular? Justify your answer.

(b) Find two singular matrices $A, B \in M_2(\mathbb{R})$ such that $A + B$ is nonsingular.

5. Determine which of the following matrices are elementary matrices.

For each elementary matrix, (i) write the corresponding elementary row operation;

(ii) find its inverse.

(a) $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ (c) $C = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $D = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (e) $E = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

6. By using matrix inversion algorithm, determine which of the following matrices are invertible. Find the inverse if it is invertible.

(a) $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

(b) $B = \begin{bmatrix} 0 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 2 & 0 \end{bmatrix}$

(c) $C = \begin{bmatrix} 5 & 5 & 0 \\ 5 & 5 & 3 \\ 5 & 5 & 1 \end{bmatrix}$

(d) $D = \begin{bmatrix} 0 & 0 & 3 & 2 \\ 1 & -3 & 2 & 0 \\ 2 & 2 & 1 & 3 \\ 0 & -1 & 5 & 2 \end{bmatrix}$

7. Let $A = \begin{bmatrix} 1 & \lambda & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

- (a) Determine all values of λ for which A is invertible.

- (b) For each value of λ in (a), find the inverse of A .

8. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ -1 & -1 & 4 \end{bmatrix}$. Show that A is invertible and find its inverse. Hence, solve the

following system of linear equations

$$\begin{aligned} x + 2y &= 1 \\ y + 3z &= 2 \\ -x - y + 4z &= 3. \end{aligned}$$

9. Let $A = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$. Verify $A^2 - 6A + 8I_3 = 0_3$. Find the inverse of A .

10. Let $A, B \in M_n(\mathbb{R})$. Suppose that A and B are row equivalent. Show that if A is invertible, then B is invertible.